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## USING CLIENT CERTIFICATES TO COMMUNICATE TRUSTED INFORMATION

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### Abstract

A device comprises: a receiver configured to receive a client certificate; a processor coupled to the receiver and configured to: authenticate the client certificate, extract, in response to the authentication, attributes from the client certificate, and create, in response to the extraction, a message comprising reformatted attributes based on the attributes, wherein the reformatted attributes can be trusted; and a transmitter coupled to the processor and configured to transmit the message. A device comprises: a processor configured to: process a client certificate comprising a certificate identifier (ID) attribute, a tenant ID attribute, and a role ID attribute, and package the client certificate in a request for a shared service; and a transmitter coupled to the processor and configured to transmit the request.

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## Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION [0001] This application is a continuation of and claims priority to U.S. patent application Ser. No. 18/402,842, filed Jan. 3, 2024, which is a continuation application of and claims priority to U.S. patent application Ser. No. 17/192,705 filed Mar. 4, 2021, which is a continuation application of and claims priority to U.S. patent application Ser. No. 16/227,147 filed Dec. 20, 2018, which is a continuation of and claims priority to U.S. patent application Ser. No. 14/211,640 filed Mar. 14, 2014, which claims priority to U.S. Provisional Application No. 61/800,548 filed Mar. 15, 2013 by Derk Norton, et al., and titled “Method of Client Authentication,” each of which is incorporated by reference. [0002] This application also incorporates by reference U.S. patent application Ser. No. 12/584,861 filed Sep. 14, 2009 by Edward R. Kelly, et al., and titled “Authentication System for Networked Computer Applications”; U.S. patent application Ser. No. 12/217,005 (now U.S. Pat. No. 8,291,490) filed Jun. 30, 2008 by Zahid N. Ahmed, et al., and titled “Tenant Life Cycle Management for a Software as a Service Platform”; U.S. patent application Ser. No. 11/533,296 (now U.S. Pat. No. 7,325,128) filed Sep. 19, 2006 by David L. Wood, et al., and titled “Log-On Service Providing Credential Level Change without Loss of Session Continuity”; U.S. patent application Ser. No. 11/224,675 (now U.S. Pat. No. 7,117,359) filed Sep. 12, 2005 by David L. Wood, et al., and titled “Default Credential Provisioning”; U.S. patent application Ser. No. 10/643,813 (now U.S. Pat. No. 6,944,761) filed Aug. 19, 2003 by David L. Wood, et al., and titled “Log-On Service Providing Credential Level Change without Loss of Session Continuity”; and U.S. patent application Ser. No. 10/431,097 filed May 7, 2003 by Derk D. Norton and titled “Client-Server Architecture Incorporating Secure Tuple Space.”

## BACKGROUND

[0003] The disclosure generally relates to computer networks employing a client-server architecture and more particularly to authentication, authorization, and auditing within such networks.

[0004] Computer networks may employ various architectures. One such architecture is a client-server architecture. In a client-server network, a client, which may be an end user, may desire to use a service on an application server. Using a client application on the client, the client may send to the application server a request for the service. In reply, the application server may send to the client a response from the service or a resource associated with the service. A gateway server may be located between the client and the application server and may function as an interface between the client and the application server. The requests and responses may follow various protocols and require authentication and authorization. In addition, the application server may audit, or log, the requests and responses.

## SUMMARY

[0005] In one embodiment, the disclosure includes a device comprising: a receiver configured to receive a client certificate; a processor coupled to the receiver and configured to: authenticate the client certificate, extract, in response to the authentication, attributes from the client certificate, and create, in response to the extraction, a message comprising reformatted attributes based on the attributes, wherein the reformatted attributes can be trusted; and a transmitter coupled to the processor and configured to transmit the message.

[0006] In another embodiment, the disclosure includes a device comprising: a processor configured to: process a client certificate comprising a certificate identifier (ID) attribute, a tenant ID attribute, and a role ID attribute, and package the client certificate in a request for a shared service; and a transmitter coupled to the processor and configured to transmit the request.

[0007] In yet another embodiment, the disclosure includes a method comprising: receiving a client certificate; authenticating the client certificate based on a signature in the client certificate; authorizing, in response to the authenticating, the client to access a shareable service; extracting, in response to the authenticating, attributes from the client certificate; and creating, in response to the extracting, a message comprising reformatted attributes based on the attributes, wherein the reformatted attributes can be trusted.

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## Description

### BRIEF DESCRIPTION OF THE DRAWINGS

[0008] FIG. 1 is a schematic diagram of a client-server network according to an embodiment of the disclosure.

[0009] FIG. 2a is a first portion of a message sequence diagram of the network described in FIG. 1 according to an embodiment of the disclosure.

[0010] FIG. 2b is a second portion of a message sequence diagram of the network described in FIG. 1 according to an embodiment of the disclosure.

[0011] FIG. 3 is a schematic diagram of the client certificate described in FIGS. 2a and 2b.

[0012] FIG. 4 is a flowchart illustrating a method of processing a client certificate according to an embodiment of the disclosure.

[0013] FIG. 5 is a schematic diagram of a computer system according to an embodiment of the disclosure.

### DETAILED DESCRIPTION

[0014] When a client requests a service from an application server, the client may request a shareable service, meaning that multiple partners are associated with the service. Shareable services may be used to, for example, manage business data and perform business transactions. First, it is important that the client be trusted before processing the client's request. Second, because shareable services are shared among multiple partners, it is important for a gateway to distinguish among the partners and to be able to maintain data privacy. As an example, the client may request a service associated with a first partner. In that case, the application server may query data for the first partner and not query data for a second partner, which might compete with the first partner, so that the application server does not divulge to the second partner data that is confidential to the first partner. The application server's access to, but distinction among, multiple partners may be referred to as multi-tenancy.

[0015] Disclosed herein are techniques for addressing the above issues. Specifically, first, the disclosure may provide for authenticating and authorizing a client, and therefore trusting the client, before processing the client's request. The request may comprise a client certificate, which may contain encoded in it at least three attributes. The attributes may include a certificate identifier (ID) attribute, a tenant ID attribute, and a role ID attribute. Second, the tenant ID attribute may identify

a specific partner and therefore prevent data confidential to that partner from being divulged to other partners. The role ID may implement role-based access control, which may refer to identifying what role the client application may fulfill while using a shareable service and prohibiting the client application from attempting to fulfill any other roles.

[0016] FIG. 1 is a schematic diagram of a client-server network **100** according to an embodiment of the disclosure. The network **100** may comprise clients **110.sub.1-m** and partner servers **170.sub.1-n** communicatively coupled to a gateway server **140** via a network **120** and through a firewall **130**. The gateway server **140** may be communicatively coupled to an application server **150**, which may, in turn, be communicatively coupled to a back end **160**. The components of the network **100** may be arranged and coupled as shown or in another suitable manner.

[0017] The clients **110.sub.1-m** may be a notebook computer, tablet computer, desktop computer, mobile telephone, or other device suitable for sending communication to, and receiving communication from, the network **120**. M may be any positive integer. The clients **110.sub.1-m** may be associated with users (not shown), who may operate the clients **110.sub.1-m** using a graphical user interface (GUI). In addition, the clients **110.sub.1-m** may comprise a client application (not shown), which may be any software application coded in any format for purposes of carrying out designated tasks based on automation or user input. The users may use the client application using the GUI. The client application may be used to access a shareable service on the application server **150**.

[0018] The network **120** may be any network suitable for allowing communication among the clients **110.sub.1-m**, the gateway server **140**, and the partner servers **170.sub.1-n**. For example, the network **120** may be the Internet or a mobile telephone network. The network **120** may allow communication along wired or wireless channels.

[0019] The firewall **130** may be a software-based or hardware-based system suitable for controlling communication to and from the server gateway **140**. The firewall **130** may control communication by applying rules to communications. The rules may be set by an administrator (not shown) via the gateway server **140**, the application server **150**, or another suitable device. The firewall **130** may include the gateway server **140**.

[0020] The gateway server **140** may be a hardware server or other device suitable for serving as an interface between the clients **110.sub.1-m** and the partner servers **170.sub.1-n** on the one hand and the application server **150** on the other hand. The gateway server **140** may translate and convert network protocols in order to allow such communication. By routing all communication to the application server **150** through the gateway server **140**, the gateway server **140** may make authentication and authorization of the clients **110.sub.1-m** simpler. In order to authenticate and authorize the clients **110.sub.1-m**, the gateway server **140** may require bi-directional Hypertext Transfer Protocol Secure (HTTPS) or other protocol authentication using mutual certificate-based Secure Sockets Layer (SSL) or Transport Layer Security (TLS), or another suitable form of authentication. HTTPS, SSL, and TLS are incorporated by reference.

[0021] The application server **150** may be a hardware server or other device suitable for storing data and providing that data to requesting clients. The application server **150** may be dedicated to providing data associated with a single service or with multiple services. When another device, for instance one of the clients **110.sub.1-m**, requests a service from the application server **150**, the application server **150** may retrieve from the back end **160** a resource associated with the service.

[0022] The back end **160** may be a device or devices suitable for storing the resources associated with the service. The back end **160** may reside within or without the application server **150**. The back end **160** may not run independently, but may instead require commands from the application server **150**. For example, the back end **160** may be a database operated using Structured Query Language (SQL), which is incorporated by reference, or any other suitable language or protocol.

[0023] The firewall **130**, the gateway server **140**, the application server **150**, and the back end **160**, or any combination of those components may be located in the network **120** or a portion of the

network **120**. Specifically, those components may be located in a cloud and operate, from the perspective of an entity associated with the application server **150**, in a cloud computing environment. In other words, those components may not be physically located where the entity associated with the application server **150** resides. The cloud may, for instance, be an Amazon® cloud.

[0024] The partner servers **170.sub.1-n** may be hardware servers or other devices suitable for sending communication to, and receiving communication from, the network **120**. **N** may be any positive integer. The partner servers **170.sub.1-n** may be associated with partners (not shown), which may be brick-and-mortar merchants such as Safeway® or Albertsons®; gift, credit, and other card issuers such as Starbucks® or Visa®; or other entities unrelated to card processing transactions. Each partner may have a specific ID associated with it. Furthermore, each partner may have multiple partner servers **170.sub.1-n** associated with it. The service described above may be a shareable service, meaning that multiple partners are associated with the service. In addition, the service may use a framework using applications that conform to representational state transfer (REST), which is incorporated by reference. Such applications may be referred to as RESTful applications.

[0025] FIG. **2a** is a first portion of a message sequence diagram **200** of the network **100** described in FIG. **1** according to an embodiment of the disclosure. At step **205**, a user (not shown) associated with the client **110.sub.1** may access a client application (not shown) using a GUI (not shown) on the client **110.sub.1**. Specifically, the user may request from the client application a service associated with the partner **170.sub.1**. The application server **150** may, however, provide the service on behalf of the partner **170.sub.1**. Accordingly, the client **110.sub.1** may begin a process of communication with the application server **150**. The communication may start, however, with mutual authentication between the client **110.sub.1** and the gateway server **140** and with the gateway server **140** authorizing the client **110.sub.1**.

[0026] Generally, steps **210-240** may comprise steps for mutual authentication between the client **110.sub.1** and the gateway server **140**. The steps may comprise messages sent between the client **110.sub.1** and the gateway server **140**, as well as among other components, such as the network **120** and the firewall **130**, between the client **110.sub.1** and the gateway server **140**. The messages may comply with SSL, TLS, or another suitable authentication protocol and be sent and received using HTTPS or another suitable communications protocol. Though steps **220-270** are shown, it may be understood that different steps may be used for SSL, TLS, or another suitable authentication protocol.

[0027] At step **210**, the client **110.sub.1** may send a request for establishment of communication with the application server **150**. For example, the client application may be an Internet browser, so the request may be the client application attempting to access a website. As another example, the client application may be a stand-alone application that directly accesses the Internet without use of a web browser. Because all communication with the application server **150** may be routed through the gateway server **140**, the gateway server **140** may handle the request. At step **215**, in response to the request at step **210**, the gateway server **140** may retrieve a server certificate either locally or from a database.

[0028] At step **220**, the gateway server **140** may send to the client **110.sub.1** the server certificate. At step **225**, the client **110.sub.1** may authenticate the gateway server **140** based on the server certificate. At step **230**, the client **110.sub.1** may retrieve a client certificate. The client **110.sub.1** may have previously received the client certificate from the partner server **170.sub.1** upon downloading the client application or at some other point. The client **110.sub.1** may store the client certificate either locally on the client **110.sub.1** or in a database accessible to the client **110.sub.1**. The client certificate may be signed by a private certificate authority associated with the application server **150**. The client certificate may have embedded within it three attributes, a certificate ID, a tenant ID, and a role ID, which are described more fully below.

[0029] At step **235**, the client **110.sub.1** may attempt to send to the application server **150** a request for a shareable service. Once again, the gateway server **140** may handle the request. The request may include the client certificate and may be related to a partner associated with the partner server **170.sub.1**. Alternatively, the client **110.sub.1** may route the request through the partner server **170.sub.1**, which may contain the client certificate.

[0030] At step **240**, the gateway server **140** may authenticate and authorize the client **110.sub.1**. The gateway server **140** may authenticate the client **110.sub.1** because it received the client certificate from the client **110.sub.1**. The client certificate may not be forged because it is signed by the private certificate authority associated with the application server **150**. The gateway server **140** may therefore authorize the client **110.sub.1** to have access to the shareable service.

[0031] FIG. **2b** is a second portion of a message sequence diagram **200** of the network **100** described in FIG. **1** according to an embodiment of the disclosure. At step **245**, after authenticating and authorizing the client **110.sub.1**, the gateway server **140** may extract the attributes from the client certificate. At step **250**, the gateway server **140** may reformat the attributes into HTTPS headers or an HTTPS body in an Extensible Markup Language (XML), JavaScript Object Notation (JSON) or other suitable message. The message may be in human readable format. At step **255**, the gateway server **140** may send to the application server **150** a request for the shareable service. The request may include the message comprising the attributes.

[0032] At step **260**, the application server **150** may audit the certificate ID, the tenant ID, and the role ID from the request for legal, billing, or other purposes. For example, if the request is related to a financial shareable service, the application server **150** may be required by law to audit the request. In addition, the application server **150** may audit the request in order to bill the client **110.sub.1** for use of the shareable service. The audit data may indicate which one of the client **100.sub.1-m**, for instance the client **100.sub.1**, made which requests. The application server **150** may store audit data either locally or on the back end **160**.

[0033] At step **265**, the application server **150** may request data from the back end **160**, which may be a resource associated with the shareable service. At step **270**, the back end **160** may send to the application server **150** the resource associated with the shareable service. At step **275**, the application server **150** may forward to the gateway server **140** the resource associated with the shareable service or a response to the request. At step **280**, the gateway server **140** may send to the client **110.sub.1** the resource associated with the shareable service or a response to the request. At step **285**, the client **110.sub.1** or the client application may use the resource or process the response.

[0034] As another example, the shareable service may be related to gift cards. A merchant may be associated with a partner server **170.sub.1**, and a user associated with the client **110.sub.1** may desire to obtain a Starbucks® gift card from the partner server **170.sub.1**. The client **110.sub.1** may be a notebook computer at a user's home. The user may access on the client **110.sub.1** a client application, which may be a web browser such as Safari®. In particular, the user may navigate to the merchant's website to obtain a Starbucks® gift card. The application server **150** may provide the shareable service. Accordingly, the partner server **170.sub.1** may contact the application server **150** through the gateway server **140**, and the partner server **170.sub.1** and the gateway server **140** may authenticate each other. After authentication, the client application may send a request to the shareable service. In other words, the client application may request a gift card or a resource related to the gift card. The application server **150** may respond with a new gift card ID, time stamps associated with the transaction, or other resources associated with the shareable service. While this example relates to gift cards, it is understood that the shareable service may be any shareable or non-shareable service provided in a client-server network such as the network **100**.

[0035] FIG. **3** is a schematic diagram of the client certificate **300** described in FIGS. **2a** and **2b**. The certificate **300** may comprise a certificate ID, or certificateId, attribute **310**; a tenant ID, or tenantId, attribute **320**; and a role ID, or roleId, attribute **330**. In a message from the client **110.sub.1** to the gateway server **140**, the certificateId attribute **310** may be referenced as a common

name (CN) attribute, the tenantId attribute **320** may be referenced as an organization (**0**) attribute, and the roleId attribute **330** may be referenced as an organizational unit (OU) attribute in the subject line of the client certificate **300**, though other associations may be made as well. In a message from the gateway server **140** to the application server **150**, the attributes may maintain their nomenclature. In addition to the certificateId attribute **310**, the tenantId attribute **320**, and the roleId attribute **330**, the client certificate **300** may comprise additional attributes that may identify other aspects of the relationship among the client **110.sub.1**, the gateway server **140**, the application server **150**, and the partner server **170.sub.1**. A private certificate authority (not shown) associated with the gateway server **140** may issue the client certificate **300**. Each partner associated with the partner servers **170.sub.1-n** may have a client certificate associated with it.

[0036] Each of the attributes may not be forged because the client certificate **300** may be signed by the private certificate authority associated with the gateway server **140**. Because the attributes may not be forged, the attributes may not need to be validated, but may instead be trusted and cannot be repudiated later by the client **110.sub.1** or other legal party. Any additional attributes similarly may be trusted and cannot be repudiated later by the client **110.sub.1** or other legal party. When the gateway server **140** extracts the attributes and forwards them to the application server **150**, the application server **150** may trust the attributes because they came from the gateway server **140**. Each attribute may be a cryptographically random globally unique identifier (GU ID), meaning that the same values may not be created twice. Because the attributes may be encoded in the client certificate **300**, they may not be changed without the client certificate **300** signature being invalidated.

[0037] The certificateId attribute **310** may be an ID that is unique to the client certificate **300**. In other words, every client certificate may have a different value for the certificateId attribute **310**. The certificateId attribute **310** may tell the gateway server **140** and the application server **150** which one of the clients **110.sub.1-m**, for instance the client **110.sub.1**, is requesting the shareable service. Before processing the tenantId attribute **320** and the roleId attribute **330**, the gateway server **140** and the application server **150** may confirm that the certificateId attribute **310** value is valid and has not been revoked. A certificateId attribute **310** value may be revoked if, for instance, the client certificate **300** has been stolen or suspected of being stolen.

[0038] The tenantId attribute **320** may be an ID that specifies which partner owns the data that the client application is attempting to access. Each partner may have a tenant ID associated with it. The tenantId attribute **320** may specify which partner data the client **110.sub.1** may access. Accordingly, the tenantId attribute **320** may ensure segregation of data associated with the partners. In other words, if the client certificate **300** comprises the tenantId attribute **320** and the tenantId attribute **320** indicates a first partner, then the first partner may not access the data of a second partner and no data associated with the second partner may be affected based on the client certificate **300**.

[0039] The roleId attribute **330** may be an ID that helps to implement role-based access control, which may refer to identifying what role the client application may fulfill while using the shareable service and prohibiting the client application from attempting to fulfill any other roles. For example, the role may be an administrator role, a general role, or another role. The types of roles may be unlimited and may dictate what data transactions the client **110.sub.1** may perform. If the roleId attribute **330** has a value indicating a general role, then the client **110.sub.1** may have limited access to data. If, however, the roleId attribute **330** has a value indicating an administrator role, then the client **110.sub.1** may have nearly unlimited access to data. There may be multiple roleId attribute **330** values associated with a single tenantId attribute **320** value in separate client certificates.

[0040] FIG. **4** is a flowchart illustrating a method **400** of processing a client certificate according to an embodiment of the disclosure. The method **400** may be implemented in the gateway server **140**. At step **410**, a client certificate may be received. The client certificate may be the client certificate **300**. At step **420**, the client certificate may be authenticated. The client certificate may be

authenticated based on a signature in the client certificate. At step **430**, access to a shareable service may be authorized. The authorization may occur in response to the authentication. The access may be authorized to one of the clients **110.sub.1-m**. At step **440**, attributes from the client certificate may be extracted. The extraction may be in response to the authentication. The attributes may be the certificateId attribute **310**, the tenantId attribute **320**, and the roleId attribute **330**. At step **450**, a message comprising reformatted attributes may be created. The creation may be in response to the extraction. The reformatted attributes may be based on the attributes and may be trusted based on the authentication.

[0041] FIG. **5** is a schematic diagram of a computer system **500** according to an embodiment of the disclosure. The system **500** may be suitable for implementing the disclosed embodiments, including the clients **110.sub.1-m**, the gateway server **140**, the application server **150**, and the partner servers **170.sub.1-n**. The system **500** may comprise a processor **510** that is in communication with memory devices, including a secondary storage **520**, a read only memory (ROM) **530**, a random access memory (RAM) **540**, input/output (I/O) devices **550**, and a transmitter/receiver **560**. Although illustrated as a single processor, the processor **510** is not so limited and may comprise multiple processors. The processor **510** may be implemented as one or more central processor unit (CPU) chips, cores (e.g., a multi-core processor), field-programmable gate arrays (FPGAs), application specific integrated circuits (ASICs), and/or digital signal processors (DSPs), and/or the processor **510** may be part of one or more ASICs. The processor **510** may be implemented using hardware or a combination of hardware and software.

[0042] The secondary storage **520** may comprise one or more disk drives or tape drives and may be used for non-volatile storage of data and as an overflow data storage device if the RAM **540** is not large enough to hold all working data. The secondary storage **520** may be used to store programs that are loaded into the RAM **540** when such programs are selected for execution. The ROM **530** may be used to store instructions and data that are read during program execution. The ROM **530** may be a non-volatile memory device that may have a small memory capacity relative to the larger memory capacity of the secondary storage **520**. The RAM **540** may be used to store volatile data and perhaps to store instructions. Access to both the ROM **530** and the RAM **540** may be faster than to the secondary storage **520**.

[0043] The transmitter/receiver **560** may serve as an output and/or input device of the system **500**. For example, if the transmitter/receiver **560** is acting as a transmitter, it may transmit data out of the system **500**. If the transmitter/receiver **560** is acting as a receiver, it may receive data into the system **500**. The transmitter/receiver **560** may take the form of modems; modem banks; Ethernet cards; universal serial bus (USB) interface cards; serial interfaces; token ring cards; fiber distributed data interface (FDDI) cards; wireless local area network (WLAN) cards; radio transceiver cards such as code division multiple access (CDMA), global system for mobile communications (GSM), long-term evolution (LTE), worldwide interoperability for microwave access (WiMAX), and/or other air interface protocol radio transceiver cards; and other well-known network devices. The transmitter/receiver **560** may enable the processor **510** to communicate with the Internet or one or more intranets. The I/O devices **550** may comprise a video monitor, a liquid crystal display (LCD), a touch screen display, or another type of video display for displaying video and may also include a video recording device for capturing video. The I/O devices **550** may also include one or more keyboards, mice, track balls, or other well-known input devices.

[0044] The ordering of steps in the various processes, data flows, and flowcharts presented are for illustration purposes and do not necessarily reflect the order that various steps must be performed. The steps may be rearranged in different orders in different embodiments to reflect the needs, desires and preferences of the entity implementing the systems. Furthermore, many steps may be performed simultaneously with other steps in some embodiments.

[0045] Also, techniques, systems, subsystems and methods described and illustrated in the various embodiments as discrete or separate may be combined or integrated with other systems, modules,



techniques, or methods without departing from the scope of the present disclosure. Other items shown or discussed as directly coupled or communicating with each other may be coupled through some interface or device, such that the items may no longer be considered directly coupled to each other but may still be indirectly coupled and in communication, whether electrically, mechanically, or otherwise with one another. Other examples of changes, substitutions, and alterations are ascertainable by one skilled in the art and could be made without departing from the spirit and scope disclosed. There has been described herein an systems and methods for providing a security code of an electronic stored-value card such that users may purchase, redeem, and/or exchange value associated with the electronic stored-value card (e.g., electronic value tokens residing in an electronic wallet). It will be apparent to those skilled in the art that modifications may be made without departing from the spirit and scope of the disclosure. The embodiments described are representative only, and are not intended to be limiting. Many variations, combinations, and modifications of the applications disclosed herein are possible and are within the scope of the disclosure. Accordingly, the scope of protection is not limited by the description set out above, but is defined by the claims which follow, that scope including all equivalents of the subject matter of the claims.

## Claims

1. A device comprising a processor implemented on one or more computer chips configured to: authenticate a client certificate, wherein the client certificate is signed by a private certificate authority associated with the device, extract, in response to the authentication, attributes from the client certificate, and create, in response to the extraction, a message comprising reformatted attributes based on the attributes, wherein the reformatted attributes comprise a certificate identifier (ID) attribute and a tenant ID attribute, wherein the certificate ID attribute identifies a client and uniquely identifies the client certificate, wherein the tenant ID attribute is associated with partner data and specifies which partner data is client accessible, and wherein the reformatted attributes can be trusted.
2. The device of claim 1, wherein the reformatted attributes further comprise a role ID attribute.
3. The device of claim 1, wherein the device is a gateway server and wherein the gateway server is configured to receive the client certificate from a client.
4. The device of claim 2, wherein the device is a gateway server and wherein the gateway server is configured to receive the client certificate from a client.
5. The device of claim 1, wherein the reformatted attributes can be trusted because of the authentication.
6. The device of claim 2, wherein the reformatted attributes can be trusted because of the authentication.
7. The device of claim 1, wherein the reformatted attributes can be trusted because the processor created the message comprising the reformatted attributes.
8. The device of claim 2, wherein the reformatted attributes can be trusted because the processor created the message comprising the reformatted attributes.
9. The device of claim 1, wherein the reformatted attributes are included in Hypertext Transfer Protocol (HTTP) or HTTP Secure (HTTPS) headers.
10. The device of claim 1, wherein the message is in an Extensible Markup Language (XML) or JavaScript Object Notation (JSON) format.
11. A device comprising a processor implemented on one or more computer chips configured to: process a client certificate's attributes to produce reformatted attributes comprising a certificate identifier (ID) attribute and a tenant ID attribute and include the client certificate in a request for a shared service, wherein the certificate identifier (ID) attribute identifies a client and uniquely identifies the device, wherein the device is the client, and wherein the tenant ID attribute specifies

which partner data is device accessible.

**12.** The device of claim 11, wherein the reformatted attributes further comprise a role ID attribute, wherein the role ID attribute helps to implement role based access control.

**13.** The device of claim 12, wherein the certificate ID attribute, the tenant ID attribute, and the role ID attribute are globally unique identifiers (GUIDs) that are unique and cannot be forged.

**14.** The device of claim 12, wherein the client certificate is signed by a private certificate authority associated with a gateway server.

**15.** The device of claim 12, wherein the request is in an Extensible Markup Language (XML) or JavaScript Object Notation (JSON) format.

**16.** A method comprising: authenticating a client certificate based on a signature in the client certificate; authorizing, in response to the authenticating, access to a shareable service; extracting, in response to the authenticating, attributes from the client certificate; and creating, in response to the extracting, a message comprising reformatted attributes based on the attributes, wherein the reformatted attributes comprise a certificate identifier (ID) attribute and a tenant ID attribute, wherein the reformatted attributes can be trusted based on the authentication, wherein the reformatted attributes are globally unique identifiers (GUIDs) that are unique and cannot be forged, wherein the certificate ID attribute uniquely identifies a device and the tenant ID attribute specifies which partner data is device accessible.

**17.** The method of claim 16, wherein the reformatted attributes further comprise a role ID attribute.

**18.** The method of claim 17, wherein the role ID attribute specifies a role of the device.

**19.** The method of claim 18, further comprising transmitting the message.

**20.** The method of claim 19, further comprising receiving, in response to the transmitting, a resource associated with the shareable service.

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